higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

14 August 2024 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used.

This question paper consists of 7 pages.

280Q1G2414

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 $5{,}24_8 + 5{,}5$

A $10{,}1_{16}$
B $10{,}1_8$
C $12{,}64_8$
D 12,4

1.2 $0001{,}00100101_{\text{bcd 8421}} + 0100{,}1000_{\text{xs3}} =$

A $2{,}C_{16}$
B $0010{,}0001_{\text{bcd 2421}}$
C 2,8
D 2,5

1.3 $18{,}3_{16} - 20{,}04_8 =$

A -12,5
B 12,5
C 12,125
D 8,125

1.4 $1{,}1_{16} \times 1{,}4_8 =$

A 4,5
B $4{,}5_8$
C 1,5
D $1{,}46_8$

1.5 $1{,}46_8 \div 1{,}8_{16}$

A 1,25
B $1{,}1_{16}$
C 1,8
D 0,25

(5 × 2) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.8) in the ANSWER BOOK.

2.1 $\overline{\overline{A} + \overline{A}B} = \overline{A}$

2.2 $\overline{A} + AB = \overline{A \bullet \overline{B}}$

2.3 $\overline{A + B + \overline{A}} = 0$

2.4 $A + \overline{B} + \overline{A} = 0$

2.5 $X + YZ + 0 = X + YZ$

2.6 $\overline{A} + B\overline{B} + \overline{A} = \overline{A}$

2.7 $\overline{A} + \overline{A}B = \overline{A} + B$

2.8 $\overline{\overline{A} + \overline{A}B} = A$

(8 × 1) **[8]**

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (3.1–3.7) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	$\overline{A \oplus B} = O/P$	A	OR
3.2	$A \oplus B = O/P$	B	NOR
3.3		C	AND
3.4		D	NAND
3.5		E	EXCLUSIVE OR
3.6		F	EXCLUSIVE NOR
3.7		G	NOT

3.3

A	B	O/P
0	0	0
0	1	0
1	0	0
1	1	1

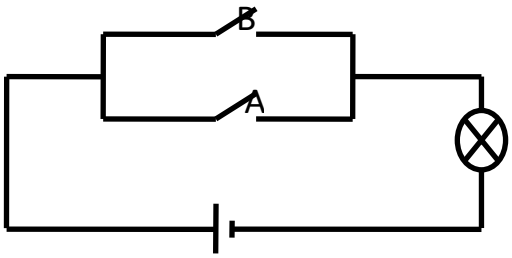
3.4

A	B	O/P
0	0	1
0	1	0
1	0	0
1	1	0

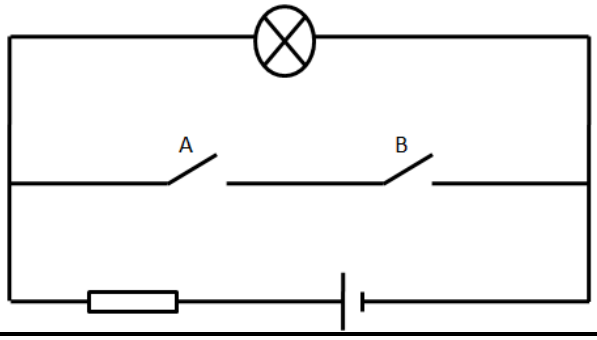
3.5

A	$\overline{A}$
0	1
1	0

3.6



3.7



(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

Complete the following calculations in the binary number system. All steps must be indicated clearly.

4.1 $101{,}01111_2 - 1000{,}1001_2$ (Make use of the 1's complement method)

4.2 $-1101{,}011_2 - 1010{,}1011_2$ (Make use of the 2's complement method)

4.3 $1010{,}0111_2 \times 100{,}101_2$

4.4 $110000{,}0100011_2 \div 1010{,}0111_2$

(4 × 5) **[20]**

QUESTION 5

Draw the circuit diagrams of the Boolean expressions below. Do not change or simplify the expressions.

5.1 $$\overline{\overline{B+\overline{A\bullet B}}\oplus C\bullet\overline{A\bullet B\bullet C}}=f$$

5.2 $$\overline{\overline{A\bullet\overline{C}}+\overline{A\bullet\overline{C}}\bullet\overline{\overline{A}\bullet B\bullet\overline{C}}}=f$$

5.3 $$\overline{A}B+\overline{B}C+A(\overline{B+C})=f$$

(3 × 5) **[15]**

QUESTION 6

Proof the Boolean statement below as 'True' or 'False' by making use of Boolean algebra and De Morgan's theorems.

If 'True', draw the circuit diagram of the statement on the right-hand side making use of only FOUR NAND gates.

$$A\oplus B=\overline{A\overline{AB}\bullet B\overline{AB}}$$

[10]

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QUESTION 7

7.1 Give the symbol and truth table for each of the following flip-flops:

7.1.1 SR

7.1.2 Clocked D-TYPE flip-flop

7.1.3 Clocked master-slave JK flip-flop

(3 × 4) (12)

7.2 Draw the circuit diagram of a four-bit series shift-right register using D-type flip-flops with a serial input and parallel outputs. Show all control and input/output points. (10)

7.3 Draw a circuit diagram of a half-adder making use of an AND gate and an EXCLUSIVE OR gate. Label the inputs and outputs clearly. Also give the truth table of the circuit diagram. (8)

[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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